

Nathan Spelts

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SUMMARY

Embedded software engineer specializing in secure firmware, low-level systems, and Rust and C development for cyber-physical systems. Passionate about reliable embedded design, secure boot, and building tools that strengthen the entire development lifecycle.

WORK EXPERIENCE

Hyster-Yale Materials Handling, Fairview, OR

Aug 2023 — Present

- Software Engineer II

- Developed embedded C firmware for safety-critical forklift microcontrollers, targeting reliability, deterministic execution, and compliance with industry safety standards.
- Implemented secure boot improvements and contributed to firmware authentication workflows, strengthening protection against tampering and unsafe OTA updates.
- Designed and built a Rust-based CAN logging and analysis tool using C FFI to interface with Vector XL hardware, enabling real-time and offline diagnostic capability.
- Improved internal debugging workflows by creating Python and Rust utilities for log parsing, firmware build artifact automation, and runtime analysis.
- Collaborated with cross-functional electrical, controls, and test engineering teams to deliver robust cyber-physical systems in industrial environments.

Seasalt.AI, Remote

Jul 2022 — May 2023

- Data Science Intern

- Improved Spanish and Bahasa Indonesian Kaldi ASR models by over 40% WER reduction through acoustic/LM tuning, pipeline optimization, and dataset refinement.
- Automated multilingual data collection and processing using Python and Bash, scraping and cleaning thousands of hours of captioned audio for model training.

EDUCATION

Georgia Institute of Technology, Remote

Aug 2025 — Present

- Master of Science in Cybersecurity – *Cyber-Physical Systems*

- Current GPA: 4.00
- Coursework focuses on embedded and systems security including secure system design, binary exploitation, network security, database security, and applied cryptography.
- Exploring resilience and hardening techniques for embedded Linux, IoT firmware, and cyber-physical systems.

Washington State University, Vancouver, WA

Graduated May 2023

- Bachelor of Science in Computer Science

- GPA: 3.9 / 4.00
- Relevant Coursework: Operating Systems, Cryptography, Embedded Systems, Data Structures, Algorithms, Software Engineering, Assembly, Statistics, and Linear Algebra.

SKILLS

Programming Languages: Rust, C, Python, C++, Lua, Bash, Powershell, Kotlin, Swift, Javascript, Java

Embedded & Systems: Secure Boot, Bootloaders, Bare-Metal ARM, RTOS Concepts, GPIO / I2C / SPI / UART / CAN, Memory-Mapped I/O, Firmware Debugging, GDB, QEMU

Tools & Technologies: Git, Docker, Kaldi, FFI (Rust/C), CI/CD Automation, Linux, LaTeX